

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA0504

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Scenario-Based Task (Medium)

Assessment Tool Weightage: 20%

QUESTIONS		
Q.No	Question	Bloom's Level
1	A university stores student and course data in a single table. Analyze the structure and identify normalization issues, then redesign it up to 3NF	. BL4 – Analyze
2	A company database contains employee and department details. Design appropriate SQL queries to retrieve employees working in multiple departments and justify your approach	. BL6 – Create
3	Given a relation with multiple functional dependencies, determine the candidate keys and check whether the relation satisfies BCNF. If not, decompose it.	BL5 – Evaluate
4	A banking system requires secure data access. Design views and constraints to ensure data security and integrity for different types of users.	BL6 – Create
5	A sales database contains redundant data leading to anomalies. Analyze the schema and apply normalization up to 3NF or BCNF to eliminate redundancy.	BL4 – Analyze
6	Given a dataset of customers and orders, write SQL queries using joins and subqueries to retrieve meaningful information such as frequent customers.	BL3 – Apply
7	A relation contains multi-valued dependencies. Analyze the structure and decompose it into 4NF, explaining the reasoning.	BL4 – Analyze

8	A database designer used improper constraints, leading to inconsistent data. Evaluate the schema and suggest appropriate integrity constraints.	BL5 – Evaluate
9	For a given relational schema, analyze the presence of join dependencies and decompose it into 5NF to remove redundancy.	BL6 – Create
10	A system uses inefficient SQL queries for data retrieval. Analyze the queries and optimize them using joins, indexing concepts, or query restructuring.	BL5 – Evaluate

Code of Conduct:

I R. Indhu(192524332)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

R. Indhu

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q1. Normalization up to 3NF

Scenario: University stores student and course details in one table.

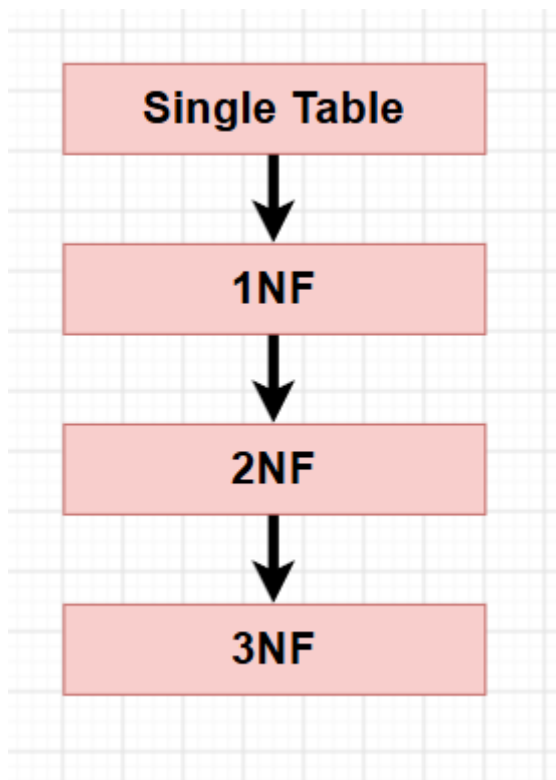
Problems:

- Data redundancy
- Update, Insert & Delete anomalies

3NF Tables:

- Student(StudentID, StudentName, Department)
- Course(CourseID, CourseName, FacultyID)
- Faculty(FacultyID, FacultyName, Phone)
- Enrollment(StudentID, CourseID)

Flow:



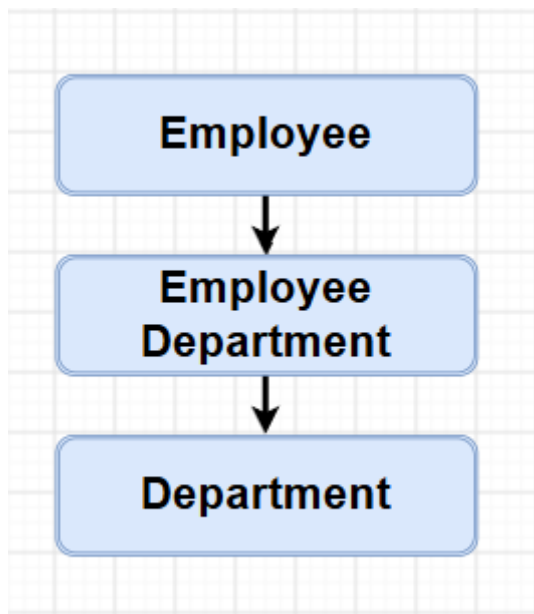
Conclusion: 3NF removes redundancy and improves consistency.

Q2. Employees in Multiple Departments

SQL Query

```
SELECT e.EmpID,e.EmpName
FROM Employee e
JOIN EmployeeDepartment ed
ON e.EmpID=ed.EmpID
GROUP BY e.EmpID,e.EmpName
HAVING COUNT(ed.DeptID)>1;
```

Diagram



Conclusion: JOIN and GROUP BY identify employees working in multiple departments.

Q3. BCNF Decomposition

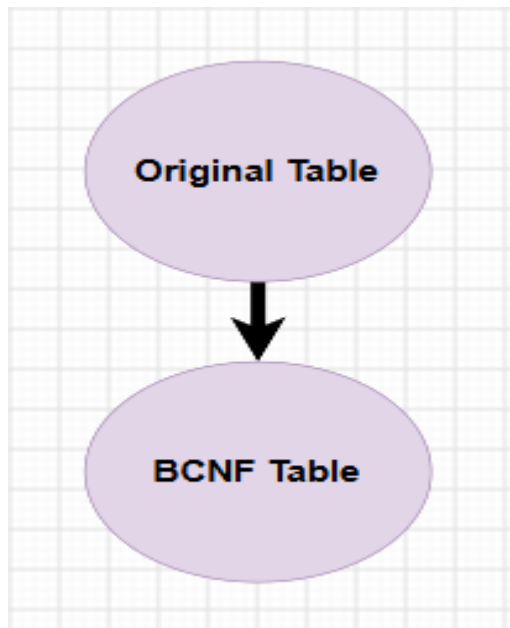
Candidate Key:

(StudentID, CourseID)

BCNF Tables

- Student(StudentID, Department)
- Course(CourseID, Faculty)
- Enrollment(StudentID, CourseID)

Diagram



Conclusion: BCNF removes dependency violations.

Q4. Banking Views and Constraints

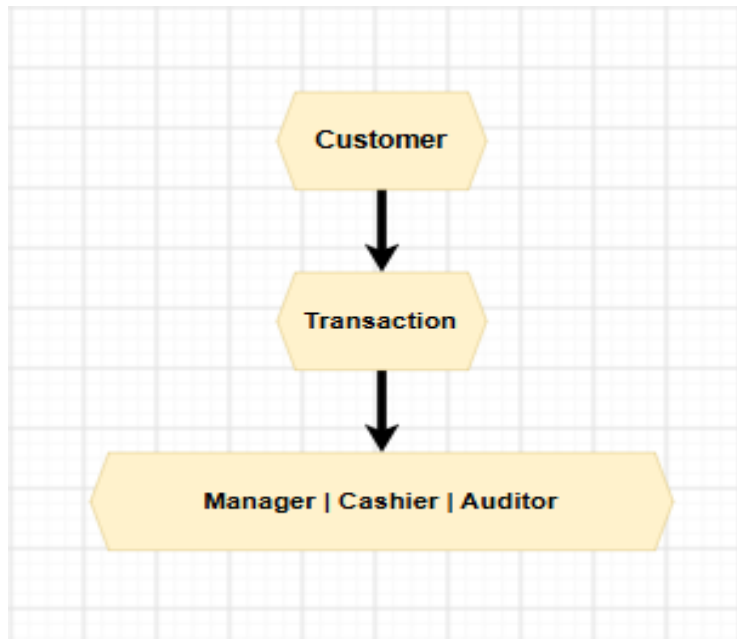
Views

- Manager_View
- Cashier_View
- Auditor_View

Constraints

- PRIMARY KEY
- FOREIGN KEY
- UNIQUE
- CHECK
- NOT NULL

Diagram



Conclusion: Views secure data and constraints maintain integrity.

Q5. Sales Database (3NF)

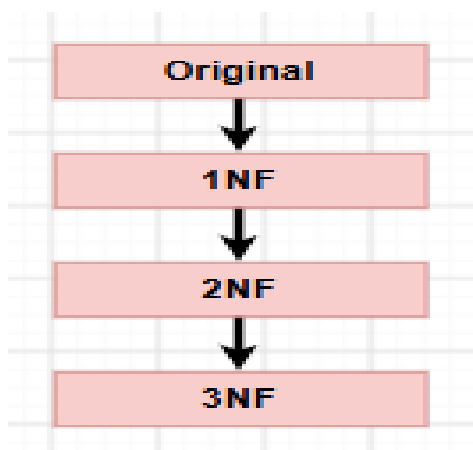
Problems

- Redundancy
- Update anomaly

Tables

- Customer
- Salesperson
- Orders

Flow



Conclusion: 3NF eliminates redundancy.

Q6. Customers and Orders

JOIN

```
SELECT C.CustomerName,  
COUNT(O.OrderID)  
FROM Customer C  
JOIN Orders O  
ON C.CustomerID=O.CustomerID  
GROUP BY C.CustomerName;
```

Subquery

```
SELECT CustomerName  
FROM Customer  
WHERE CustomerID IN  
(SELECT CustomerID  
FROM Orders  
GROUP BY CustomerID  
HAVING COUNT(*)>2);
```

Conclusion: Identifies frequent customers.

Q7. 4NF Decomposition

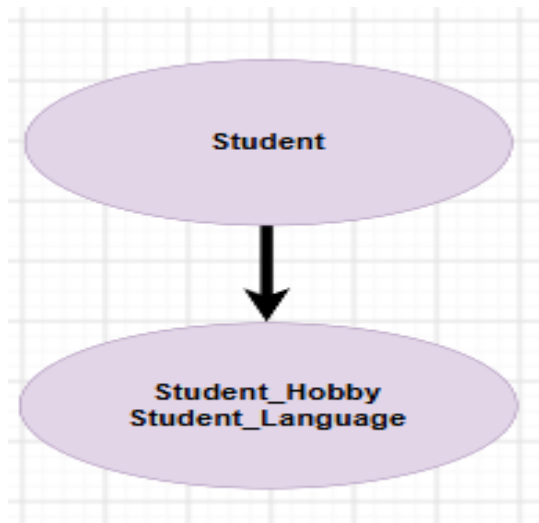
Relation

Student(StudentID,Hobby,Language)

Tables

- Student_Hobby
- Student_Language

Diagram



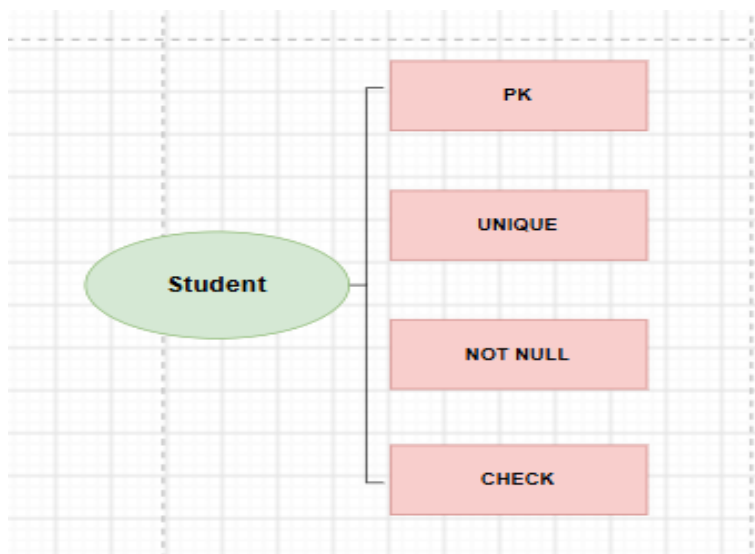
Conclusion: 4NF removes multivalued dependencies.

Q8. Integrity Constraints

Constraints

- PRIMARY KEY
- UNIQUE
- NOT NULL
- CHECK

Diagram



Conclusion: Prevents duplicate and invalid data.

Q9. 5NF Decomposition

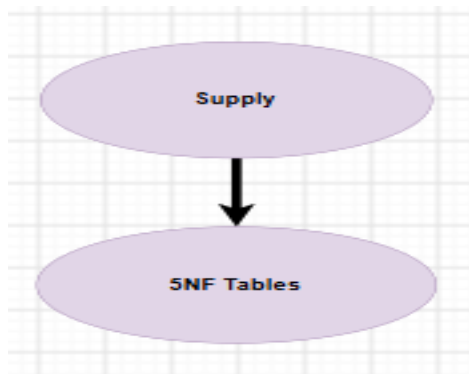
Relation

Supply(Supplier, Part, Project)

Tables

- Supplier_Part
- Supplier_Project
- Part_Project

Diagram



Conclusion: 5NF removes join redundancy.

Q10. SQL Query Optimization

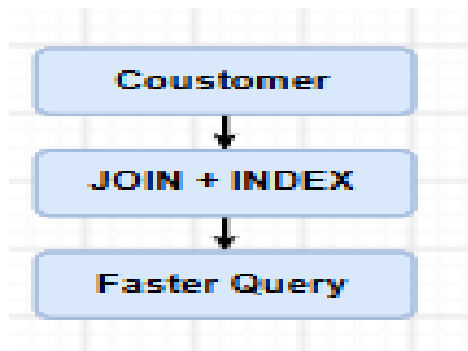
Optimized Query

```
SELECT DISTINCT C.CustomerID,C.CustomerName
FROM Customer C
JOIN Orders O
ON C.CustomerID=O.CustomerID;
```

Index

```
CREATE INDEX idx_customer
ON Orders(CustomerID);
```

Diagram



Conclusion: JOIN and INDEX improve query performance and reduce execution time.